

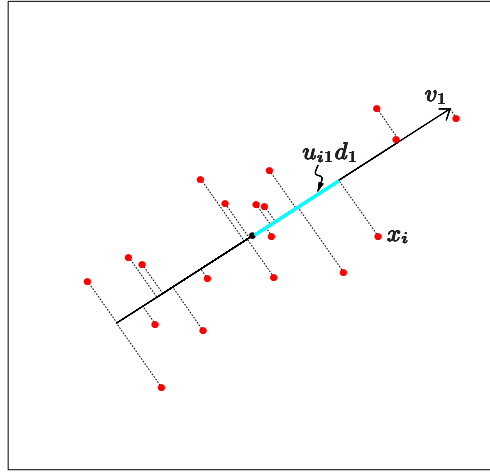


FIGURE 14.20. The first linear principal component of a set of data. The line minimizes the total squared distance from each point to its orthogonal projection onto the line.

In this application the authors have developed a “zoom” feature, which allows one to interact with the map in order to get more detail. The final level of zooming retrieves the actual news articles, which can then be read.

14.5 Principal Components, Curves and Surfaces

Principal components are discussed in Sections 3.4.1, where they shed light on the shrinkage mechanism of ridge regression. Principal components are a sequence of projections of the data, mutually uncorrelated and ordered in variance. In the next section we present principal components as linear manifolds approximating a set of N points $x_i \in \mathbb{R}^p$. We then present some nonlinear generalizations in Section 14.5.2. Other recent proposals for nonlinear approximating manifolds are discussed in Section 14.9.

14.5.1 Principal Components

The principal components of a set of data in $\mathbb{R}^p$ provide a sequence of best linear approximations to that data, of all ranks $q \leq p$.

Denote the observations by $x_1, x_2, \dots, x_N$, and consider the rank- q linear model for representing them

$$f(\lambda) = \mu + \mathbf{V}_q \lambda, \quad (14.49)$$

where μ is a location vector in $\mathbb{R}^p$, $\mathbf{V}_q$ is a $p \times q$ matrix with q orthogonal unit vectors as columns, and λ is a q vector of parameters. This is the parametric representation of an affine hyperplane of rank q . Figures 14.20 and 14.21 illustrate for $q = 1$ and $q = 2$, respectively. Fitting such a model to the data by least squares amounts to minimizing the *reconstruction error*

$$\min_{\mu, \{\lambda_i\}, \mathbf{V}_q} \sum_{i=1}^N \|x_i - \mu - \mathbf{V}_q \lambda_i\|^2. \quad (14.50)$$

We can partially optimize for μ and the λ_i (Exercise 14.7) to obtain

$$\hat{\mu} = \bar{x}, \quad (14.51)$$

$$\hat{\lambda}_i = \mathbf{V}_q^T (x_i - \bar{x}). \quad (14.52)$$

This leaves us to find the orthogonal matrix $\mathbf{V}_q$:

$$\min_{\mathbf{V}_q} \sum_{i=1}^N \|(x_i - \bar{x}) - \mathbf{V}_q \mathbf{V}_q^T (x_i - \bar{x})\|^2. \quad (14.53)$$

For convenience we assume that $\bar{x} = 0$ (otherwise we simply replace the observations by their centered versions $\tilde{x}_i = x_i - \bar{x}$). The $p \times p$ matrix $\mathbf{H}_q = \mathbf{V}_q \mathbf{V}_q^T$ is a *projection matrix*, and maps each point x_i onto its rank- q reconstruction $\mathbf{H}_q x_i$, the orthogonal projection of x_i onto the subspace spanned by the columns of $\mathbf{V}_q$. The solution can be expressed as follows. Stack the (centered) observations into the rows of an $N \times p$ matrix $\mathbf{X}$. We construct the *singular value decomposition* of $\mathbf{X}$:

$$\mathbf{X} = \mathbf{U} \mathbf{D} \mathbf{V}^T. \quad (14.54)$$

This is a standard decomposition in numerical analysis, and many algorithms exist for its computation (Golub and Van Loan, 1983, for example). Here $\mathbf{U}$ is an $N \times p$ orthogonal matrix ($\mathbf{U}^T \mathbf{U} = \mathbf{I}_p$) whose columns $\mathbf{u}_j$ are called the *left singular vectors*; $\mathbf{V}$ is a $p \times p$ orthogonal matrix ($\mathbf{V}^T \mathbf{V} = \mathbf{I}_p$) with columns v_j called the *right singular vectors*, and $\mathbf{D}$ is a $p \times p$ diagonal matrix, with diagonal elements $d_1 \geq d_2 \geq \dots \geq d_p \geq 0$ known as the *singular values*. For each rank q , the solution $\mathbf{V}_q$ to (14.53) consists of the first q columns of $\mathbf{V}$. The columns of $\mathbf{U} \mathbf{D}$ are called the principal components of $\mathbf{X}$ (see Section 3.5.1). The N optimal $\hat{\lambda}_i$ in (14.52) are given by the first q principal components (the N rows of the $N \times q$ matrix $\mathbf{U}_q \mathbf{D}_q$).

The one-dimensional principal component line in $\mathbb{R}^2$ is illustrated in Figure 14.20. For each data point x_i , there is a closest point on the line, given by $u_{i1} d_1 v_1$. Here v_1 is the direction of the line and $\hat{\lambda}_i = u_{i1} d_1$ measures distance along the line from the origin. Similarly Figure 14.21 shows the

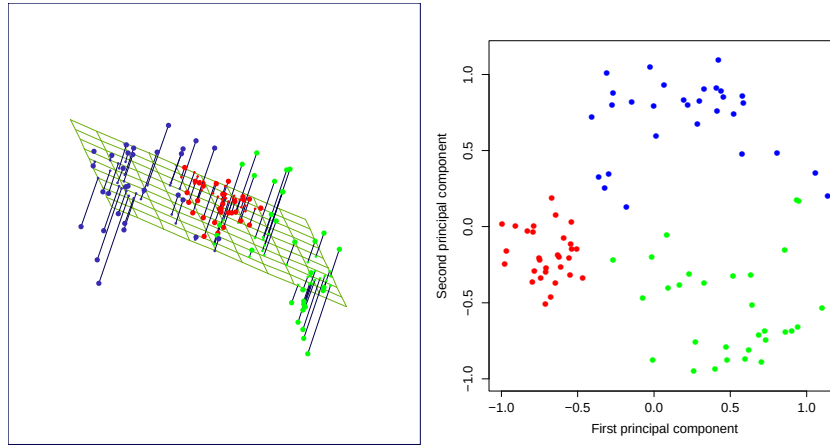


FIGURE 14.21. The best rank-two linear approximation to the half-sphere data. The right panel shows the projected points with coordinates given by $\mathbf{U}_2\mathbf{D}_2$, the first two principal components of the data.

two-dimensional principal component surface fit to the half-sphere data (left panel). The right panel shows the projection of the data onto the first two principal components. This projection was the basis for the initial configuration for the SOM method shown earlier. The procedure is quite successful at separating the clusters. Since the half-sphere is nonlinear, a nonlinear projection will do a better job, and this is the topic of the next section.

Principal components have many other nice properties, for example, the linear combination $\mathbf{X}v_1$ has the highest variance among all linear combinations of the features; $\mathbf{X}v_2$ has the highest variance among all linear combinations satisfying v_2 orthogonal to v_1 , and so on.

Example: Handwritten Digits

Principal components are a useful tool for dimension reduction and compression. We illustrate this feature on the handwritten digits data described in Chapter 1. Figure 14.22 shows a sample of 130 handwritten 3's, each a digitized 16×16 grayscale image, from a total of 658 such 3's. We see considerable variation in writing styles, character thickness and orientation. We consider these images as points x_i in $\mathbb{R}^{256}$, and compute their principal components via the SVD (14.54).

Figure 14.23 shows the first two principal components of these data. For each of these first two principal components u_{i1} and u_{i2} , we computed the 5%, 25%, 50%, 75% and 95% quantile points, and used them to define the rectangular grid superimposed on the plot. The circled points indicate

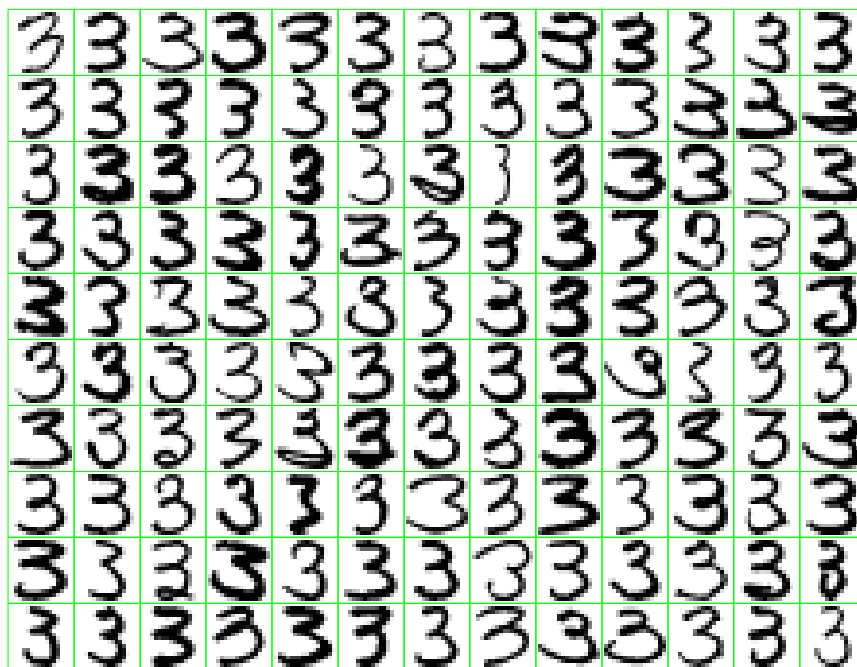


FIGURE 14.22. A sample of 130 handwritten 3's shows a variety of writing styles.

those images close to the vertices of the grid, where the distance measure focuses mainly on these projected coordinates, but gives some weight to the components in the orthogonal subspace. The right plot shows the images corresponding to these circled points. This allows us to visualize the nature of the first two principal components. We see that the v_1 (horizontal movement) mainly accounts for the lengthening of the lower tail of the three, while v_2 (vertical movement) accounts for character thickness. In terms of the parametrized model (14.49), this two-component model has the form

$$\begin{aligned}\hat{f}(\lambda) &= \bar{x} + \lambda_1 v_1 + \lambda_2 v_2 \\ &= \boxed{3} + \lambda_1 \cdot \boxed{3} + \lambda_2 \cdot \boxed{3}.\end{aligned}\quad (14.55)$$

Here we have displayed the first two principal component directions, v_1 and v_2 , as images. Although there are a possible 256 principal components, approximately 50 account for 90% of the variation in the threes, 12 account for 63%. Figure 14.24 compares the singular values to those obtained for equivalent uncorrelated data, obtained by randomly scrambling each column of $\mathbf{X}$. The pixels in a digitized image are inherently correlated, and since these are all the same digit the correlations are even stronger.

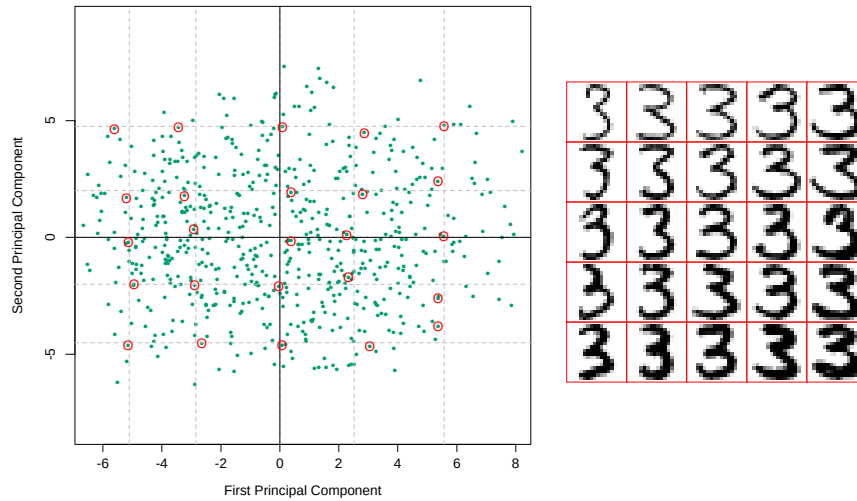


FIGURE 14.23. (Left panel:) the first two principal components of the handwritten threes. The circled points are the closest projected images to the vertices of a grid, defined by the marginal quantiles of the principal components. (Right panel:) The images corresponding to the circled points. These show the nature of the first two principal components.

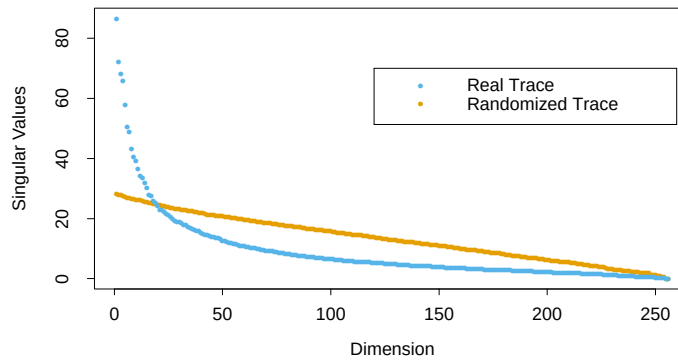


FIGURE 14.24. The 256 singular values for the digitized threes, compared to those for a randomized version of the data (each column of $\mathbf{X}$ was scrambled).

A relatively small subset of the principal components serve as excellent lower-dimensional features for representing the high-dimensional data.

Example: Procrustes Transformations and Shape Averaging

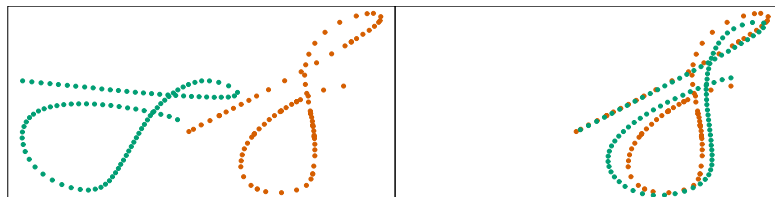


FIGURE 14.25. (Left panel:) Two different digitized handwritten *S*s, each represented by 96 corresponding points in $\mathbb{R}^2$. The green *S* has been deliberately rotated and translated for visual effect. (Right panel:) A Procrustes transformation applies a translation and rotation to best match up the two set of points.

Figure 14.25 represents two sets of points, the orange and green, in the same plot. In this instance these points represent two digitized versions of a handwritten *S*, extracted from the signature of a subject “Suresh.” Figure 14.26 shows the entire signatures from which these were extracted (third and fourth panels). The signatures are recorded dynamically using touch-screen devices, familiar sights in modern supermarkets. There are $N = 96$ points representing each *S*, which we denote by the $N \times 2$ matrices $\mathbf{X}_1$ and $\mathbf{X}_2$. There is a correspondence between the points—the i th rows of $\mathbf{X}_1$ and $\mathbf{X}_2$ are meant to represent the same positions along the two *S*’s. In the language of morphometrics, these points represent *landmarks* on the two objects. How one finds such corresponding landmarks is in general difficult and subject specific. In this particular case we used *dynamic time warping* of the speed signal along each signature (Hastie et al., 1992), but will not go into details here.

In the right panel we have applied a translation and rotation to the green points so as best to match the orange—a so-called *Procrustes*³ transformation (Mardia et al., 1979, for example).

Consider the problem

$$\min_{\mu, \mathbf{R}} \|\mathbf{X}_2 - (\mathbf{X}_1 \mathbf{R} + \mathbf{1}\mu^T)\|_F, \quad (14.56)$$

³Procrustes was an African bandit in Greek mythology, who stretched or squashed his visitors to fit his iron bed (eventually killing them).

with $\mathbf{X}_1$ and $\mathbf{X}_2$ both $N \times p$ matrices of corresponding points, $\mathbf{R}$ an orthonormal $p \times p$ matrix⁴, and μ a p -vector of location coordinates. Here $\|\mathbf{X}\|_F^2 = \text{trace}(\mathbf{X}^T \mathbf{X})$ is the squared *Frobenius* matrix norm.

Let $\bar{x}_1$ and $\bar{x}_2$ be the column mean vectors of the matrices, and $\tilde{\mathbf{X}}_1$ and $\tilde{\mathbf{X}}_2$ be the versions of these matrices with the means removed. Consider the SVD $\tilde{\mathbf{X}}_1^T \tilde{\mathbf{X}}_2 = \mathbf{U} \mathbf{D} \mathbf{V}^T$. Then the solution to (14.56) is given by (Exercise 14.8)

$$\begin{aligned}\hat{\mathbf{R}} &= \mathbf{U} \mathbf{V}^T \\ \hat{\mu} &= \bar{x}_2 - \hat{\mathbf{R}} \bar{x}_1,\end{aligned}\tag{14.57}$$

and the minimal distance is referred to as the *Procrustes distance*. From the form of the solution, we can center each matrix at its column centroid, and then ignore location completely. Hereafter we assume this is the case.

The *Procrustes distance with scaling* solves a slightly more general problem,

$$\min_{\beta, \mathbf{R}} \|\mathbf{X}_2 - \beta \mathbf{X}_1 \mathbf{R}\|_F, \tag{14.58}$$

where $\beta > 0$ is a positive scalar. The solution for $\mathbf{R}$ is as before, with $\hat{\beta} = \text{trace}(\mathbf{D}) / \|\mathbf{X}_1\|_F^2$.

Related to Procrustes distance is the *Procrustes average* of a collection of L shapes, which solves the problem

$$\min_{\{\mathbf{R}_\ell\}_1^L, \mathbf{M}} \sum_{\ell=1}^L \|\mathbf{X}_\ell \mathbf{R}_\ell - \mathbf{M}\|_F^2; \tag{14.59}$$

that is, find the shape $\mathbf{M}$ closest in average squared Procrustes distance to all the shapes. This is solved by a simple alternating algorithm:

0. Initialize $\mathbf{M} = \mathbf{X}_1$ (for example).
1. Solve the L Procrustes rotation problems with $\mathbf{M}$ fixed, yielding $\mathbf{X}'_\ell \leftarrow \mathbf{X}_\ell \hat{\mathbf{R}}_\ell$.
2. Let $\mathbf{M} \leftarrow \frac{1}{L} \sum_{\ell=1}^L \mathbf{X}'_\ell$.

Steps 1. and 2. are repeated until the criterion (14.59) converges.

Figure 14.26 shows a simple example with three shapes. Note that we can only expect a solution up to a rotation; alternatively, we can impose a constraint, such as that $\mathbf{M}$ be upper-triangular, to force uniqueness. One can easily incorporate scaling in the definition (14.59); see Exercise 14.9.

Most generally we can define the *affine-invariant* average of a set of shapes via

⁴To simplify matters, we consider only orthogonal matrices which include reflections as well as rotations [the $O(p)$ group]; although reflections are unlikely here, these methods can be restricted further to allow only rotations [$SO(p)$ group].

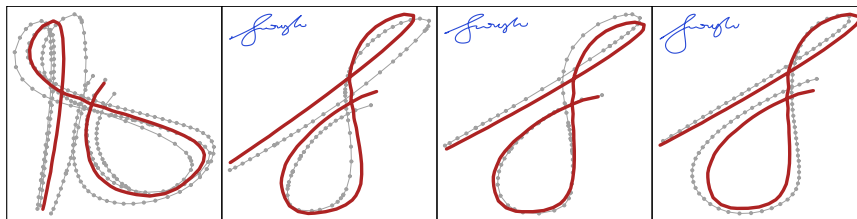


FIGURE 14.26. The Procrustes average of three versions of the leading S in Suresh's signatures. The left panel shows the preshape average, with each of the shapes $\mathbf{X}'_\ell$ in preshape space superimposed. The right three panels map the preshape $\mathbf{M}$ separately to match each of the original S 's.

$$\min_{\{\mathbf{A}_\ell\}_{\ell=1}^L, \mathbf{M}} \sum_{\ell=1}^L \|\mathbf{X}_\ell \mathbf{A}_\ell - \mathbf{M}\|_F^2, \quad (14.60)$$

where the $\mathbf{A}_\ell$ are any $p \times p$ nonsingular matrices. Here we require a standardization, such as $\mathbf{M}^T \mathbf{M} = \mathbf{I}$, to avoid a trivial solution. The solution is attractive, and can be computed without iteration (Exercise 14.10):

1. Let $\mathbf{H}_\ell = \mathbf{X}_\ell (\mathbf{X}_\ell^T \mathbf{X}_\ell)^{-1} \mathbf{X}_\ell^T$ be the rank- p projection matrix defined by $\mathbf{X}_\ell$.
2. $\mathbf{M}$ is the $N \times p$ matrix formed from the p largest eigenvectors of $\bar{\mathbf{H}} = \frac{1}{L} \sum_{\ell=1}^L \mathbf{H}_\ell$.

14.5.2 Principal Curves and Surfaces

Principal curves generalize the principal component line, providing a smooth one-dimensional curved approximation to a set of data points in $\mathbb{R}^p$. A principal surface is more general, providing a curved manifold approximation of dimension 2 or more.

We will first define principal curves for random variables $X \in \mathbb{R}^p$, and then move to the finite data case. Let $f(\lambda)$ be a parameterized smooth curve in $\mathbb{R}^p$. Hence $f(\lambda)$ is a vector function with p coordinates, each a smooth function of the single parameter λ . The parameter λ can be chosen, for example, to be arc-length along the curve from some fixed origin. For each data value x , let $\lambda_f(x)$ define the closest point on the curve to x . Then $f(\lambda)$ is called a principal curve for the distribution of the random vector X if

$$f(\lambda) = E(X | \lambda_f(X) = \lambda). \quad (14.61)$$

This says $f(\lambda)$ is the average of all data points that project to it, that is, the points for which it is “responsible.” This is also known as a *self-consistency* property. Although in practice, continuous multivariate distributions have infinitely many principal curves (Duchamp and Stuetzle, 1996), we are

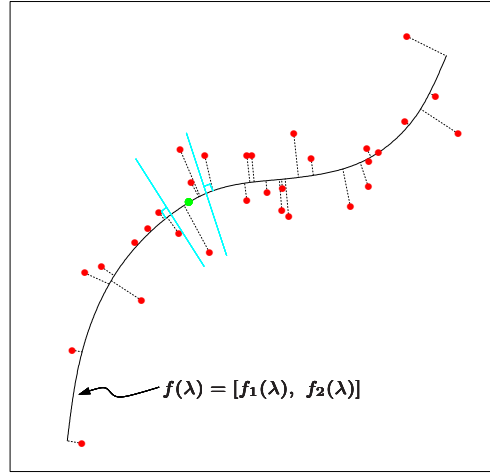


FIGURE 14.27. The principal curve of a set of data. Each point on the curve is the average of all data points that project there.

interested mainly in the smooth ones. A principal curve is illustrated in Figure 14.27.

Principal points are an interesting related concept. Consider a set of k prototypes and for each point x in the support of a distribution, identify the closest prototype, that is, the prototype that is responsible for it. This induces a partition of the feature space into so-called Voronoi regions. The set of k points that minimize the expected distance from X to its prototype are called the principal points of the distribution. Each principal point is self-consistent, in that it equals the mean of X in its Voronoi region. For example, with $k = 1$, the principal point of a circular normal distribution is the mean vector; with $k = 2$ they are a pair of points symmetrically placed on a ray through the mean vector. Principal points are the distributional analogs of centroids found by K -means clustering. Principal curves can be viewed as $k = \infty$ principal points, but constrained to lie on a smooth curve, in a similar way that a SOM constrains K -means cluster centers to fall on a smooth manifold.

To find a principal curve $f(\lambda)$ of a distribution, we consider its coordinate functions $f(\lambda) = [f_1(\lambda), f_2(\lambda), \dots, f_p(\lambda)]$ and let $X^T = (X_1, X_2, \dots, X_p)$. Consider the following alternating steps:

$$\begin{aligned} \text{(a)} \quad \hat{f}_j(\lambda) &\leftarrow E(X_j | \lambda(X) = \lambda); \quad j = 1, 2, \dots, p, \\ \text{(b)} \quad \hat{\lambda}_f(x) &\leftarrow \operatorname{argmin}_{\lambda'} \|x - \hat{f}(\lambda')\|^2. \end{aligned} \quad (14.62)$$

The first equation fixes λ and enforces the self-consistency requirement (14.61). The second equation fixes the curve and finds the closest point on

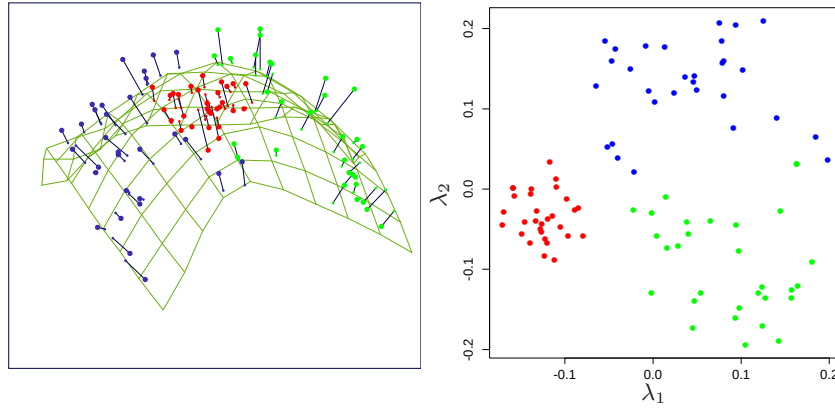


FIGURE 14.28. *Principal surface fit to half-sphere data. (Left panel:) fitted two-dimensional surface. (Right panel:) projections of data points onto the surface, resulting in coordinates $\hat{\lambda}_1, \hat{\lambda}_2$.*

the curve to each data point. With finite data, the principal curve algorithm starts with the linear principal component, and iterates the two steps in (14.62) until convergence. A scatterplot smoother is used to estimate the conditional expectations in step (a) by smoothing each X_j as a function of the arc-length $\hat{\lambda}(X)$, and the projection in (b) is done for each of the observed data points. Proving convergence in general is difficult, but one can show that if a linear least squares fit is used for the scatterplot smoothing, then the procedure converges to the first linear principal component, and is equivalent to the *power method* for finding the largest eigenvector of a matrix.

Principal surfaces have exactly the same form as principal curves, but are of higher dimension. The mostly commonly used is the two-dimensional principal surface, with coordinate functions

$$f(\lambda_1, \lambda_2) = [f_1(\lambda_1, \lambda_2), \dots, f_p(\lambda_1, \lambda_2)].$$

The estimates in step (a) above are obtained from two-dimensional surface smoothers. Principal surfaces of dimension greater than two are rarely used, since the visualization aspect is less attractive, as is smoothing in high dimensions.

Figure 14.28 shows the result of a principal surface fit to the half-sphere data. Plotted are the data points as a function of the estimated nonlinear coordinates $\hat{\lambda}_1(x_i), \hat{\lambda}_2(x_i)$. The class separation is evident.

Principal surfaces are very similar to self-organizing maps. If we use a kernel surface smoother to estimate each coordinate function $f_j(\lambda_1, \lambda_2)$, this has the same form as the batch version of SOMs (14.48). The SOM weights w_k are just the weights in the kernel. There is a difference, however:

the principal surface estimates a separate prototype $f(\lambda_1(x_i), \lambda_2(x_i))$ for each data point x_i , while the SOM shares a smaller number of prototypes for all data points. As a result, the SOM and principal surface will agree only as the number of SOM prototypes grows very large.

There also is a conceptual difference between the two. Principal surfaces provide a smooth parameterization of the entire manifold in terms of its coordinate functions, while SOMs are discrete and produce only the estimated prototypes for approximating the data. The smooth parameterization in principal surfaces preserves distance locally: in Figure 14.28 it reveals that the red cluster is tighter than the green or blue clusters. In simple examples the estimates coordinate functions themselves can be informative: see Exercise 14.13.

14.5.3 Spectral Clustering

Traditional clustering methods like K -means use a spherical or elliptical metric to group data points. Hence they will not work well when the clusters are non-convex, such as the concentric circles in the top left panel of Figure 14.29. Spectral clustering is a generalization of standard clustering methods, and is designed for these situations. It has close connections with the local multidimensional-scaling techniques (Section 14.9) that generalize MDS.

The starting point is a $N \times N$ matrix of pairwise similarities $s_{ii'} \geq 0$ between all observation pairs. We represent the observations in an undirected *similarity graph* $G = \langle V, E \rangle$. The N vertices v_i represent the observations, and pairs of vertices are connected by an edge if their similarity is positive (or exceeds some threshold). The edges are weighted by the $s_{ii'}$. Clustering is now rephrased as a graph-partition problem, where we identify connected components with clusters. We wish to partition the graph, such that edges between different groups have low weight, and within a group have high weight. The idea in spectral clustering is to construct similarity graphs that represent the local neighborhood relationships between observations.

To make things more concrete, consider a set of N points $x_i \in \mathbb{R}^p$, and let $d_{ii'}$ be the Euclidean distance between x_i and $x_{i'}$. We will use as similarity matrix the radial-kernel gram matrix; that is, $s_{ii'} = \exp(-d_{ii'}^2/c)$, where $c > 0$ is a scale parameter.

There are many ways to define a similarity matrix and its associated similarity graph that reflect local behavior. The most popular is the *mutual K -nearest-neighbor graph*. Define $\mathcal{N}_K$ to be the symmetric set of nearby pairs of points; specifically a pair (i, i') is in $\mathcal{N}_K$ if point i is among the K -nearest neighbors of i' , or vice-versa. Then we connect all symmetric nearest neighbors, and give them edge weight $w_{ii'} = s_{ii'}$; otherwise the edge weight is zero. Equivalently we set to zero all the pairwise similarities not in $\mathcal{N}_K$, and draw the graph for this modified similarity matrix.